Data_Scientist — Step 4

Master Machine Learning

Detailed Learning Notes • CareerCompass

1. Learning Objective

Understand how to formulate, train, evaluate and compare machine-learning models as part of a data-science workflow.

2. Core Concepts — Detailed Breakdown

Problem formulation

Define target, features, unit of prediction, prediction horizon and evaluation metric.

Supervised learning

Learn regression and classification, including linear/logistic regression, trees, ensembles and nearest-neighbor methods.

Unsupervised learning

Understand clustering and dimensionality reduction and their exploratory role.

Generalization

Understand train/validation/test splits, cross-validation, overfitting, regularization and hyperparameter tuning.

Evaluation

Choose metrics based on the decision problem. Classification may require precision, recall, F1, ROC-AUC or PR-AUC; regression may use MAE, MSE/RMSE or R^2.

Interpretation

Use feature importance or other explanation techniques carefully and distinguish association from causal effect.

3. Recommended Learning Workflow

- Build a simple baseline.
- Create a preprocessing pipeline.
- Use cross-validation for model comparison.
- Tune only using training/validation data.
- Evaluate once on the protected test set.
- Inspect errors and subgroup behavior.

5. Hands-on Project / Practice

Build a predictive project with at least three models. Compare them using cross-validation, select an appropriate metric, perform error analysis, and explain why the final model is preferable beyond its raw score.

6. Common Mistakes & Pitfalls

- Leakage during preprocessing.
- Optimizing on the test set.
- Using accuracy on imbalanced data.
- Ignoring calibration when probabilities drive decisions.
- Assuming feature importance is causal.

7. Completion Checklist

- I can define a target and metric.
- I can build a pipeline.
- I can compare models with cross-validation.
- I can explain overfitting and regularization.
- I can interpret evaluation results responsibly.

8. Interview / Assessment Questions

- What is cross-validation?
- Why is the test set protected?
- What is overfitting?
- When is PR-AUC useful?
- What is data leakage?

9. Recommended References

Scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html

Machine Learning in 2024: https://www.youtube.com/watch?v=bmmQA8A-yUA

Kaggle Learn: https://www.kaggle.com/learn

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.